Pancreatic Cancer Detection Using Machine Learning Algorithms

A Seminar Report Presented By,

Shivani Prasad

A123008

Department of Computer Science and Engineering

International Institute of Information Technology Bhubaneswar

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Contents

UNDERTAKING

I declare that the work presented in this seminar titled “Pancreatic Cancer Detection,” submitted to the Department, International Institute of Information Technology, Bhubaneswar, for the award of the Masters of Technology degree in Computer Science Engineering, is my original work. I have not plagiarised or submitted the same work for the award of any other degree. If this undertaking is found to be incorrect, I accept that my degree may be unconditionally withdrawn.

IIIT Bhubaneswar

Shivani Prasad

A12008

Abstract

Pancreatic cancer detection poses a significant challenge due to its often asymptomatic early stages and deep-seated location within the body. Early detection is crucial for effective treatment and improved outcomes. This study explores the use of machine learning (ML) models for early pancreatic cancer detection using a dataset comprising various clinical and biomarker features. The dataset underwent preprocessing steps, including handling missing values and label encoding. We employed a pipeline incorporating standardization and selected ML classifiers to train and evaluate the models. The top-performing classifiers were assessed based on accuracy, precision, recall, and F1-score metrics. Confusion matrices were generated to visualize classifier performance. The study aimed to identify the most effective ML model for pancreatic cancer detection. Results suggest promising performance across various classifiers, with the best classifier achieving an accuracy of 88 per cent. The findings underscore the potential of ML techniques in enhancing pancreatic cancer detection, thereby facilitating early intervention and improving patient outcomes.

Chapter 1

Introduction

Pancreatic cancer stands as a formidable adversary in the oncological domain, characterized by its dauntingly high mortality rates and a mere 10% five-year survival rate [1]. It is projected to ascend as the second leading cause of cancer-related deaths globally, attributable in part to the less advanced treatment modalities available compared to other cancer types, compounded by the escalating prevalence of pancreatic cancer worldwide. Early detection of this malignancy poses a significant challenge owing to its often asymptomatic nature and the deep-seated location of the pancreas within the body, resulting in a majority of cases being diagnosed at advanced stages, thus severely limiting treatment options and contributing to a dismal prognosis [1].

However, amidst these grim realities, a beacon of hope shines forth in the form of machine learning (ML) technology. ML has emerged as a potent tool with the potential to revolutionize the early detection of pancreatic cancer by analyzing extensive datasets encompassing diverse patient information, ranging from clinical records to imaging studies and molecular biomarkers. ML algorithms exhibit the capability to discern subtle patterns and trends within these datasets, thereby facilitating the identification of pancreatic cancer at its nascent stages when interventions are most efficacious [2].

The integration of ML into the realm of pancreatic cancer detection holds promise for improving outcomes and saving lives. By harnessing the analytical prowess of ML algorithms, healthcare providers can glean insights to make more informed decisions, customise treatment strategies to suit individual patient profiles, and ultimately enhance survival rates [2]. However, the realization of ML's potential in pancreatic cancer detection is not devoid of challenges, including ethical considerations, data privacy concerns, and the imperative for stringent validation [3].

In this context, this study endeavours to explore the convergence of ML and pancreatic cancer detection, to elucidate the methodologies, challenges, and opportunities inherent in leveraging ML technology to combat this lethal disease. Through a comprehensive examination of contemporary research and advancements in the field, we aspire to shed light on the transformative potential of ML in enhancing the diagnosis of pancreatic cancer and improving patient outcomes [4].

1.1 Background

Pancreatic cancer is notorious for its high fatality rates, emphasizing the critical need for early detection to improve patient survival rates. Conventional screening methods often fail to identify pancreatic cancer at its nascent stages, resulting in limited treatment options and dismal prognoses. Recent studies underscore the challenges in early pancreatic cancer detection and highlight the importance of innovative approaches [5], [6]. Insights from medical institutions also emphasize the potential of AI-driven innovations in enhancing early identification methods for pancreatic cancer, providing further motivation for research in this area [7].

1.2 Motivation

Detecting pancreatic cancer in early stages is challenging because the pancreas is located deep inside the body. During regular physical exams, healthcare providers cannot see or feel early tumours. Typically, individuals do not experience any noticeable symptoms until the cancer has grown significantly or has spread to other organs. If someone has a family history of pancreatic cancer, their risk of developing the disease may be higher than usual. Early detection of the disease at an early stage can greatly enhance the effectiveness of treatment.

1.3 Aim

The aim is to develop a system that creates a sophisticated system to detect pancreatic cancer. Enable early detection for better treatment outcomes and to save lives. Use the latest technologies for precise identification and diagnosis in the early stages. Revolutionize pancreatic cancer diagnosis and treatment to improve quality of life.

Chapter 2

Literature Survey

In recent years, research in cancer detection and diagnosis has witnessed significant advancements, driven by the integration of machine learning techniques. A variety of studies have explored innovative approaches to enhance cancer classification accuracy, leveraging hybrid systems, deep learning methods, and ensemble techniques. For instance, the study presented in "A Hybrid System Based on Genetic Algorithm and Adaptive Neuro-Fuzzy Inference System for Cancer Classification" [8] introduces a novel hybrid model combining genetic algorithms and adaptive neuro-fuzzy inference systems for cancer classification, demonstrating improved accuracy in detecting various cancer types. Additionally, the integration of deep learning techniques with transcriptomics data, as demonstrated in "Integrating Deep Learning and Transcriptomics Reveals the Genetic Landscape of Pancreatic Cancer" [9], offers valuable insights into the genetic landscape of pancreatic cancer, paving the way for more precise diagnostic approaches. Furthermore, advancements in deep learning-based approaches, such as those discussed in "Deep Learning-Based Pancreatic Cancer Detection and Diagnosis Using Computed Tomography Images" [10] and "Application of Deep Learning Techniques in Pancreatic Cancer Detection Using Microarray Data" [11], underscore the potential of deep learning models in improving diagnostic accuracy and streamlining cancer detection processes. Moreover, hybrid approaches, such as the one presented in "A Hybrid Approach for Cancer Diagnosis Based on Ensemble Learning and Genetic Algorithm" [12], showcase the effectiveness of combining ensemble learning techniques with genetic algorithms to enhance cancer diagnostic capabilities. These collective efforts highlight the ongoing endeavors to leverage machine learning and computational methods for advancing cancer detection, diagnosis, and ultimately improving patient outcomes.

Chapter 3

Dataset And Proposed Model

3.1 Dataset

The original dataset contains 590 entries and 12 columns, including patient cohort, sample origin, age, sex, diagnosis, stage, and various biomarkers such as plasma_CA19_9, creatinine, LYVE1, REG1B, TFF1, and REG1A. The data types range from float64 to int64, with one object type for patient cohort. This dataset offers a comprehensive array of demographic, clinical, and biomarker information essential for pancreatic cancer analysis

```
Original Data Information:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 590 entries, 0 to 589
Data columns (total 12 columns):
 #   Column          Non-Null Count  Dtype  
---  ------          --------------  -----  
 0   patient_cohort  590 non-null    object 
 1   sample_origin   590 non-null    int64  
 2   age             590 non-null    int64  
 3   sex             590 non-null    int64  
 4   diagnosis       590 non-null    int64  
 5   stage           590 non-null    int64  
 6   plasma_CA19_9   590 non-null    float64
 7   creatinine      590 non-null    float64
 8   LYVE1           590 non-null    float64
 9   REG1B           590 non-null    float64
 10  TFF1            590 non-null    float64
 11  REG1A           590 non-null    float64
dtypes: float64(6), int64(5), object(1)
memory usage: 55.4+ KB
None
```

Figure 1: Original Data Information

After preprocessing, the dataset was reduced to 395 entries and 10 columns. Summary statistics show the mean, standard deviation, minimum, 25th percentile, median, 75th percentile, and maximum values for each column in the preprocessed data. The preprocessing steps involved scaling numerical features and encoding categorical variables, ensuring the data is ready for machine learning model training.

```
Preprocessed Data Information:
Shape of preprocessed data: (395, 10)


Summary Statistics of Preprocessed Data:
                  0             1             2           3             4  \
count  3.950000e+02  3.950000e+02  3.950000e+02  395.000000  3.950000e+02
mean   5.621382e-17  1.708900e-16 -5.846238e-17    0.000000 -3.597685e-17
std    1.001268e+00  1.001268e+00  1.001268e+00    1.001268  1.001268e+00
min   -6.273450e-01 -2.639349e+00 -9.874209e-01   -0.717811 -3.196740e-01
25%   -6.273450e-01 -7.353061e-01 -9.874209e-01   -0.717811 -3.078854e-01
50%   -6.273450e-01  5.804520e-02 -9.874209e-01   -0.717811 -1.060974e-02
75%    1.002927e+00  6.927262e-01  1.012739e+00    1.255753 -1.060974e-02
max    2.633198e+00  2.358764e+00  1.012739e+00    1.913608  1.433134e+01

                  5             6             7             8             9
count  3.950000e+02  3.950000e+02  3.950000e+02  3.950000e+02  3.950000e+02
mean   9.668778e-17  4.047395e-17 -6.745659e-18 -4.497106e-17 -7.195369e-17
std    1.001268e+00  1.001268e+00  1.001268e+00  1.001268e+00  1.001268e+00
min   -1.265032e+00 -9.031775e-01 -6.113084e-01 -5.917719e-01 -7.404492e-01
25%   -7.775217e-01 -8.306769e-01 -5.527384e-01 -5.300687e-01 -5.406028e-01
50%   -2.067356e-01 -4.315477e-01 -3.979935e-01 -3.323012e-01 -8.107264e-03
75%    5.045783e-01  6.190245e-01  4.937502e-02  1.400613e-01 -8.107264e-03
max    4.963302e+00  5.998768e+00  5.853053e+00  1.207179e+01  7.955648e+00
```

Figure 2: Preprocessed Data Information

The biomarkers used in the dataset after preprocessing include plasma_CA19_9, creatinine, LYVE1, REG1B, TFF1, and REG1A. These biomarkers play a crucial role in pancreatic cancer detection and were standardized during preprocessing for further analysis. The data was sourced from Kaggle, and preprocessing was performed to enhance its usability for machine learning algorithms.

3.2 Proposed Model

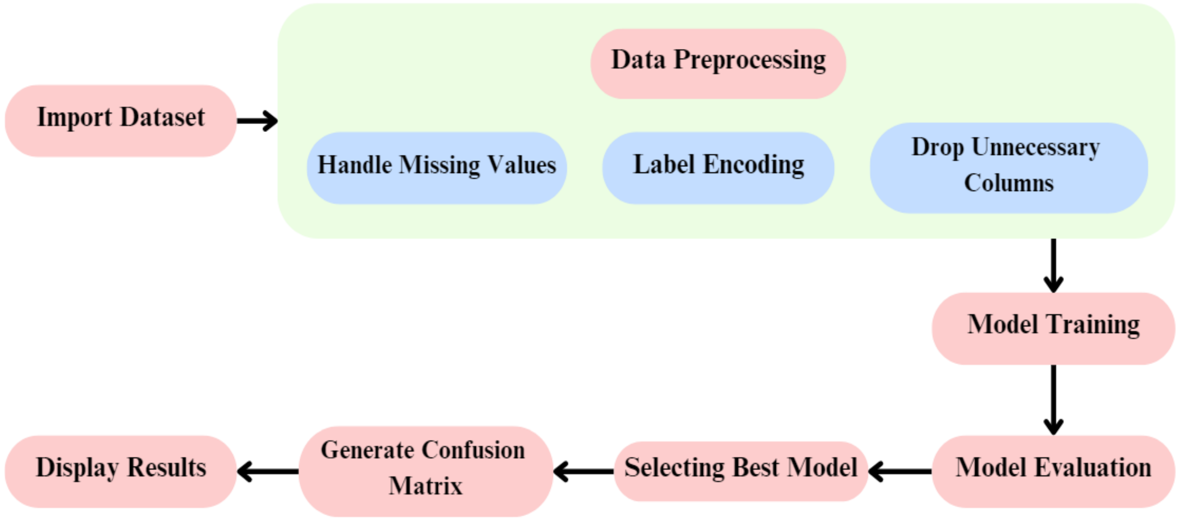

Figure 3: A Strategic Roadmap

The process begins with importing the dataset, which contains relevant information for pancreatic cancer detection. The data then undergoes preprocessing, including handling missing values, label encoding categorical variables, and dropping unnecessary columns to ensure compatibility and effectiveness for model training. Following preprocessing, the dataset is split into training and testing sets to facilitate model training and evaluation. Machine learning algorithms are applied to the training data to build predictive models, which are subsequently evaluated using separate validation data. The best-performing model is selected based on predefined criteria such as accuracy. A confusion matrix is generated to visualize the model's performance, displaying key metrics such as true positives, true negatives, false positives, and false negatives. Finally, the results of the model evaluation process, including performance metrics and relevant information, are displayed to stakeholders.

Chapter 4

Algorithms Used

4.1 HistGradientBoostingClassifier

Histogram Gradient Boosting, an extension of the gradient boosting algorithm, revolutionizes ensemble learning by introducing histogram-based decision trees. Unlike traditional gradient boosting, which builds trees level-wise, histogram gradient boosting constructs trees in a depth-wise manner, dramatically reducing training time. By leveraging histogram binning techniques, it accelerates computation by grouping data into discrete bins. This innovation enhances efficiency without compromising predictive accuracy, making it ideal for large-scale datasets like those encountered in medical research. Despite its sequential nature, histogram gradient boosting offers superior performance in structured predictive modeling tasks, making it a preferred choice for various applications due to its balance of speed and accuracy [13].

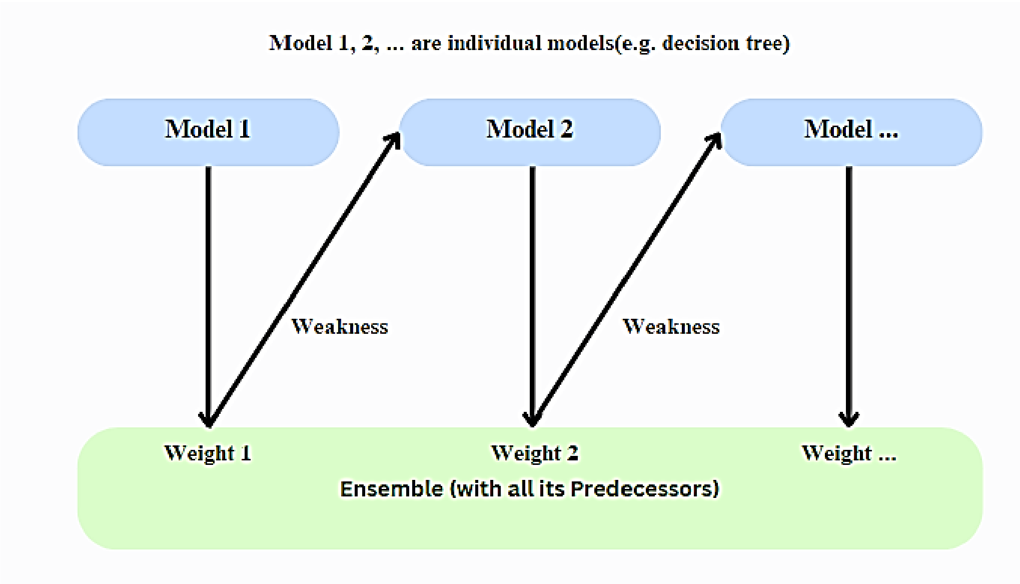

Figure 4: Histogram Gradient Boosting

4.2 ExtraTreesClassifier

The ExtraTreesClassifier, enhances ensemble learning by introducing extreme randomization during tree construction. Unlike traditional decision trees or random forests, ExtraTreesClassifier randomly selects candidate splits at each node without evaluating the best possible split. This randomized approach significantly reduces computation time, making it highly efficient for large datasets. By introducing additional randomness, ExtraTreesClassifier effectively mitigates overfitting and improves generalization performance. This method is particularly useful in high-dimensional and noisy data settings, where it achieves competitive performance with reduced computational resources. Overall, ExtraTreesClassifier offers a robust and scalable solution for various machine learning tasks [14].

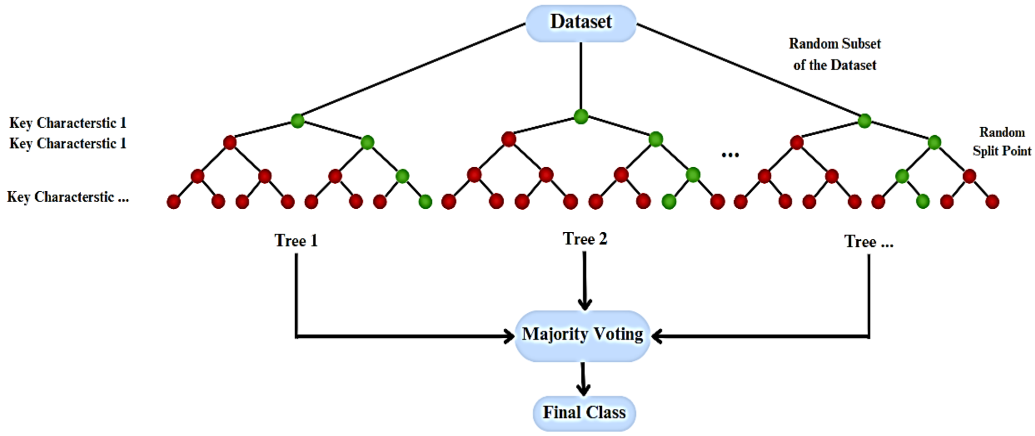

Figure 5: Extra Trees Classifier

4.3 BaggingClassifier

BaggingClassifier, a machine learning technique, harnesses the concept of bagging predictors. It aggregates predictions from multiple base classifiers trained on different subsets of the training data, aiming to improve overall performance. By averaging or voting on these predictions, BaggingClassifier reduces variance and mitigates overfitting, enhancing the model's generalization capability. This approach is particularly effective in situations where individual classifiers may produce highly variable predictions or suffer from instability. BaggingClassifier has proven to be a robust and widely used ensemble learning algorithm in various applications, including classification and regression tasks [15].

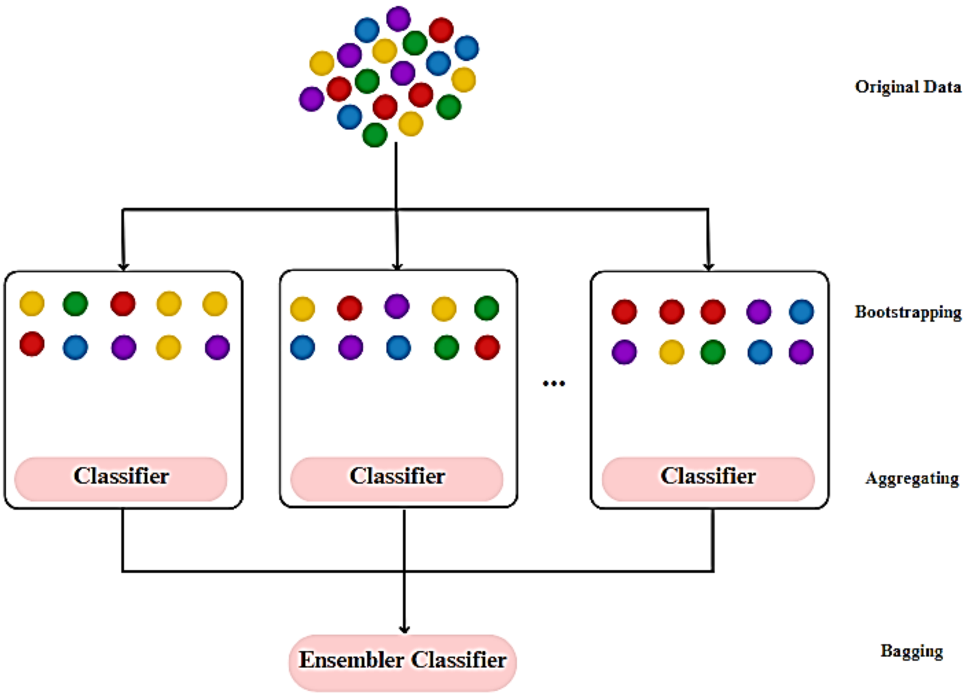

Figure 6: Bagging Classifier

4.4 CalibratedClassifierCV

CalibratedClassifierCV leverages the technique of transforming classifier scores into accurate multiclass probability estimates. This method refines the output probabilities of base classifiers to align them with true class proportions. By iteratively adjusting the probability estimates, CalibratedClassifierCV ensures reliable and well-calibrated predictions. This approach is crucial for applications requiring precise risk assessment or diagnostic predictions, as it enhances the interpretability and trustworthiness of the model's output [16].

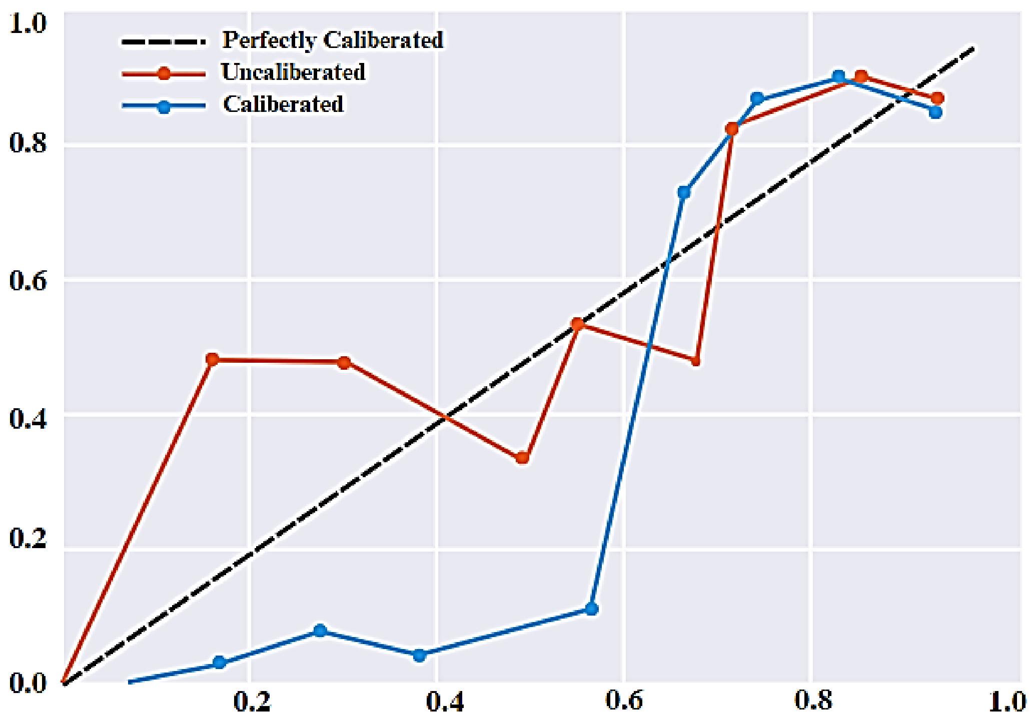

Figure 7: Calibrated Classifier

4.5 GaussianProcessClassifier

GaussianProcessClassifier, as outlined by Chris Williams in "Gaussian Processes for Machine Learning," employs Gaussian processes for classification tasks. It leverages the principles of Bayesian inference to model the posterior distribution over functions, allowing for probabilistic predictions. By capturing uncertainty and modeling complex relationships between inputs and outputs, GaussianProcessClassifier offers a flexible and powerful framework for classification. It's particularly beneficial in scenarios where data is limited or noisy, as it provides a principled approach to handle uncertainty and make reliable predictions. This method has found applications in various fields, including healthcare, finance, and natural language processing [17].

Chapter 5

Result And Analysis

5.1 Confusion Matrix

A confusion matrix is a valuable tool for measuring the effectiveness of a machine learning model, particularly in classification tasks. Essentially, it's a table that provides a summary of a classification algorithm's performance by comparing the actual and predicted classes for each instance in a dataset.

The confusion matrix has rows and columns representing the actual and predicted classes, respectively. It helps visualize the performance of a classifier by breaking down the counts of true positive (TP), true negative (TN), false positive (FP), and false negative (FN) predictions.

- **True Positive (TP):** The number of instances correctly predicted as positive.
- **True Negative (TN):** The number of instances correctly predicted as negative.
- **False Positive (FP):** The number of instances incorrectly predicted as positive when actually negative.
- **False Negative (FN):** The number of instances incorrectly predicted as negative when actually positive.

The confusion matrix allows practitioners to calculate various performance metrics such as accuracy, precision, recall, and F1-score, providing insights into the strengths and weaknesses of the classification model.

5.2 Accuracy

Accuracy is a measure of the overall correctness of a model. It calculates the ratio of correctly predicted instances to the total number of instances in the dataset. In other words, accuracy tells us how often the classifier is correct. It is expressed as a percentage.

$$\text{Accuracy} = \frac{\text{Total Number of Predictions}}{\text{Number of Correct Predictions}}$$

5.3 Precision

Precision, also known as positive predictive value, measures the proportion of true positive predictions among all positive predictions made by the classifier. It focuses on the accuracy of positive predictions, indicating how many of the predicted positive instances are actually positive.

$$\text{Precision} = \frac{\text{True Positives}}{\text{True Positives} + \text{False Positives}}$$

5.4 Recall

Recall, also known as true positive rate or sensitivity, measures the proportion of true positive predictions among all actual positive instances in the dataset. It focuses on the ability of the classifier to find all the positive instances, indicating how many of the actual positive instances are correctly predicted.

$$\text{Recall} = \frac{\text{True Positives}}{\text{True Positives} + \text{False Negatives}}$$

5.5 F1-Score

F1-score is the harmonic mean of precision and recall. It provides a balance between precision and recall, especially when the class distribution is imbalanced. F1-score ranges from 0 to 1, where a score closer to 1 indicates better model performance. It is a useful metric for evaluating the overall performance of a classifier, considering both false positives and false negatives.

$$\text{F1 Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

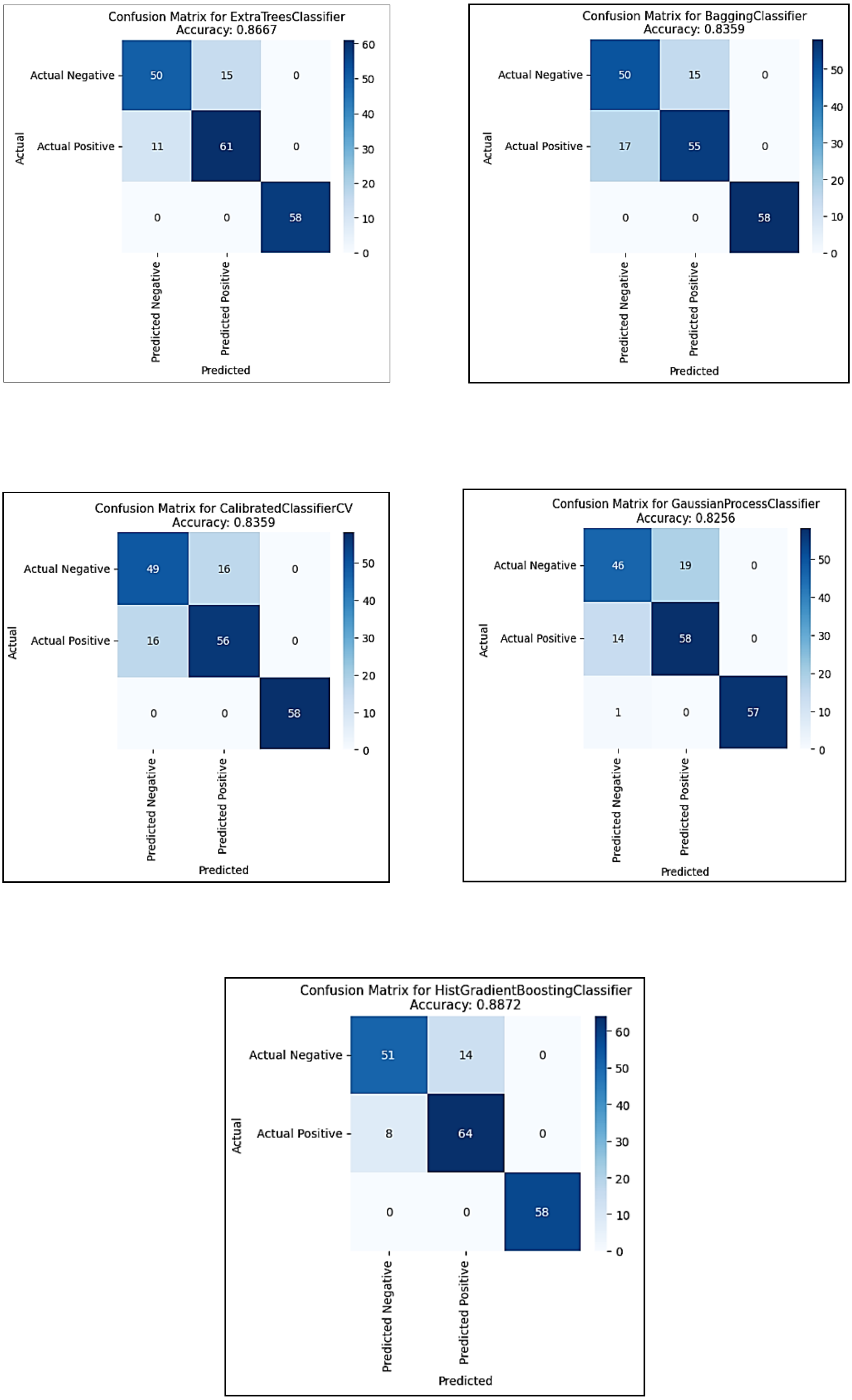

Figure 8: Confusion Matrices for all the classifiers

	Classifier	Accuracy	Precision	Recall	F1-score
0	HistGradientBoostingClassifier	0.887179	0.887662	0.887179	0.886913
1	ExtraTreesClassifier	0.882051	0.882867	0.882051	0.881671
2	BaggingClassifier	0.835897	0.836302	0.835897	0.835985
3	CalibratedClassifierCV	0.835897	0.835897	0.835897	0.835897
4	GaussianProcessClassifier	0.825641	0.826924	0.825641	0.825691

Figure 9: All the classifiers and their calculations

```
Best Classifier: HistGradientBoostingClassifier, Accuracy: 0.8871794871794871

Summary of the Best Classifier:
Metric      Value
----------  ------------------------------
Classifier  HistGradientBoostingClassifier
Accuracy    0.8872
Precision   0.8877
Recall      0.8872
F1-score    0.8869
```

Figure 10: Best Classifier

The **HistGradientBoostingClassifier** demonstrates the highest accuracy among the classifiers evaluated. With an accuracy of 0.8872, it performs well in correctly predicting the diagnosis of pancreatic cancer cases based on the dataset features.

Chapter 6

Conclusion And Future Work

6.1 Conclusion

The study addresses the pressing issue of pancreatic cancer detection, which poses significant challenges due to its often asymptomatic early stages and deep-seated location within the body. The focus of the study is to explore the efficacy of machine learning algorithms in accurately predicting pancreatic cancer diagnosis using a dataset comprising various clinical and biomarker features. The methods employed include feature selection, preprocessing techniques such as handling missing values and label encoding, and the utilization of a pipeline incorporating standardization and selected machine learning classifiers. Key findings reveal that the HistGradientBoostingClassifier emerged as the standout performer among the evaluated classifiers, achieving the highest accuracy along with balanced precision and recall. This underscores its effectiveness in accurately predicting pancreatic cancer diagnosis within the dataset. Moreover, the selected features and preprocessing techniques played a pivotal role in contributing to the model's success. These findings highlight the dataset's suitability for leveraging machine learning algorithms in pancreatic cancer prediction. Moving forward, continued analysis and refinement of these models hold promise for further enhancing performance and potentially yielding valuable insights for both diagnosis and treatment of cancer.

6.2 Future Work

In the future, pancreatic cancer detection through machine learning is poised for significant advancements. By refining feature engineering and embracing deep learning techniques like CNNs and RNNs, we can uncover nuanced patterns for more precise predictions. Integrating diverse data sources and prioritizing model interpretability will bolster trust in clinical applications. Real-time decision support and personalized medicine approaches promise to streamline diagnosis and optimize treatment. However, ethical considerations and rigorous validation across patient cohorts are essential for responsible implementation. Collaboration between researchers, clinicians, and policymakers will be critical for translating these innovations into impactful clinical practice.

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